Supplement to When Does Verification Help? Recovery-Path Regularization for Strategy-Aware Valuation

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This supplement reports additional fixed-design diagnostics and the matched stronger-gap comparison. It preserves the distinction between new training, new diagnostics from existing models, and the earlier four-stage audit. It does not establish new-market effectiveness or repair the retained strict cross-machine numerical failure. All models and per-listing artifacts remain local; released tables contain source-, seed- or batch-level aggregates.

S1 Data construction and operational assumptions

Sample flow. Historical construction receipts bind the input versions and every split checksum. Joint filtering precedes non-price fingerprint deduplication; individual exclusion flags overlap and must not be added. The counts below reconcile all retained records, including calibration and test partitions. Only train and validation enter the reported fixed-cohort experiments; calibration and test are not newly evaluated here. Sealed test labels were hashed as bytes for integrity, not parsed for this revision. JUCars retains used-condition records with positive finite JOD price and

Table S1: Construction counts. Filtered counts are joint exclusions; deduplication follows filtering.

Source	Raw	Filtered	Duplicates	Retained	Used train/eval
Synthetic	7500	0	0	7500	5000/1000
JUCars	10921	2117	46	8758	4774/1000
MUCars	101896	28892	1693	71311	5000/1000

Table S2: Partitions before label-independent experimental truncation. Synthetic validation denotes its separately generated evaluation population.

Source	Train	Validation	Calibration	Test
Synthetic	6000	1500	0	0
JUCars	4774	1374	1343	1267
MUCars	39244	10672	10696	10699

model year 1980–2026. MUCars excludes records labeled New, restricts advertised MAD price to [5,000,2,000,000], and model year to 1980–2025. These are inherited quality filters, not a claim

of full-source representativeness. Split buckets use the first eight hexadecimal digits of a non-price fingerprint hash modulo 100: 0–54 train, 55–69 validation, 70–84 calibration, 85–99 test. No original advertisement ID or verified vehicle identity is available. Identical non-price fingerprints may over-collapse different vehicles. Training and evaluation are separately hash-ranked before the 5,000/1,000 caps. The same fixed populations and batches are shared by fitting seeds.

Field mapping and missingness. JUCars uses its mileage_km, transmission, fuel_type and feat_body_condition fields. MUCars maps Mileage, Gearbox, Fuel and Condition to these four fields; mileage bands use numeric midpoints, or the stated bound for a one-ended band. Missingness means unavailable in the processed source record. Its actual cause is unknown: it may reflect collection, inapplicability or earlier concealment. Unrecoverability is a simulation stipulation, not an empirical diagnosis of those causes. Numeric inputs use training medians, missing indicators and scaling; mileage is log-transformed. Categorical missing tokens and training-fit one-hot encodings retain unknown-category handling. Targets, IDs, source indices and declared reveal aliases are excluded from predictor inputs. Sellers can conceal at most two available fields, each record receives

Table S3: Unavailable fields in the actual experimental cohorts (percent of records). Full retained and pre-cap split counts are also released in field-missingness.csv.

Source	Cohort	n	Mileage	Gearbox	Fuel	Condition
Synthetic	training	5000	3.880	4.280	4.120	4.160
Synthetic	evaluation	1000	3.700	3.800	3.800	3.500
JUCars	training	4774	20.381	0.021	0.021	0.063
JUCars	evaluation	1000	19.900	0.000	0.000	0.100
MUCars	training	5000	0.000	0.000	0.000	3.500
MUCars	evaluation	1000	0.000	0.000	0.000	2.800

Table S4: Distribution of legal report counts (percent of records). With m available fields, the count is $1 + m + \binom{m}{2}$; legal concealed-field recovery edges number m^2.

Source	Cohort	1	2	4	7	11
Synthetic	training	0.00	0.04	0.96	14.40	84.60
Synthetic	evaluation	0.00	0.20	0.70	12.80	86.30
JUCars	training	0.00	0.00	0.04	20.40	79.56
JUCars	evaluation	0.00	0.00	0.00	20.00	80.00
MUCars	training	0.00	0.00	0.00	3.50	96.50
MUCars	evaluation	0.00	0.00	0.00	2.80	97.20

at most one unit-cost query, failed queries spend capacity, and the primary concealment cost is zero. Extra absence or positive-cost tests of earlier models are not relabeled as RPL robustness tests.

Synthetic generator. Draw make $M \sim U\{0,\dots,4\}$, variant $V \sim U\{0,\dots,3\}$ and age $A \sim U\{1,\dots,18\}$. Mileage is $K = \max(100, A\exp Z)$ with $Z \sim N(\log 15000, .4^2)$. Condition $C \in \{0,1,2,3\}$ has probabilities $(.12,.28,.42,.18)$. Automatic transmission $T \sim$ Bernoulli$(.45+.06\mathbf{1}[M>2])$, diesel $D \sim$ Bernoulli$(.4)$ and fiscal power $P = 4+M+2V+U\{0,1,2\}$.

Log target is

$$y = 10.3 + .23M + .1V - .072A - .20\log(1 + K/50000) \\ + .12C + .08T + .04D + .02P - .061[C = 0, A > 10] + \epsilon, \quad \epsilon \sim N(0, .1^2).$$

Year is $2024 - A$, model identifies make/variant, and location is uniform over three labels. Four action fields are independently made unavailable with probability .04 after target generation. Shared latent attributes create feature correlations. Seeds 73101/73102 generate 6,000/1,500 train/evaluation records. This is a stipulated generator, not inferred seller behavior.

Training schedules by experiment family. The inherited backbone has 250 boosting iterations and no fitted neural residual when branches start. The earlier manuscript's six-initial-epoch and two-three-epoch description was incorrect; the bound production config and training receipts specify twelve residual epochs as shown below. No model or result was changed to match the corrected description.

Table S5: Residual update schedules. Optimizer state resets at each update call; epochs and report-search sweeps are different quantities.

Experiment family	Residual calls	Training response
Original feedback factorial	Two calls of six epochs	One start/two sweeps, fixed epoch scores
Original static no-feedback	One call of twelve epochs	No query
Matched RPL, gap controls and loss ablation	Two calls of six epochs	No query; common report maximization
Search-strength intervention	Two calls of six epochs	One/two versus four/twenty; epoch/batch refresh
Parameter/stability variants	Explicit per-variant settings	Declared variant; no transfer of scope
RPL by five acquisition families	No new price-model fitting	Reused predictors; new evaluation responses

S2 Ten-seed recovery geometry, quotes and budget interaction

All-edge diagnostics. We infer all eleven report-state quotes from each of sixty existing deployment models (three sources, ten fitting seeds, $\lambda = 0, 1$ at $\beta = .5$). For every legal concealed-field recovery edge, compute $[g(T(s,a)) - g(s)]_+$ with $g(s) = [q(s) - q(H)]_+$. Reported rates use threshold 10^{-8} and all legal edges as denominator. Quantiles below first include zero increments within each model and are then averaged across seeds. They are not pooled-population quantiles or confidence limits. Full per-model values, including maxima, are in edge-60-models.csv. Every edge ending at H has zero penalty because $g(H) = 0$ and $g(s) \geq 0$. Thus the regularizer targets intermediate recovery states. An improved positive-gain graph need not lower the raw quote on every recovery edge and does not establish truthfulness or a fixed honest anchor.

Quote and query outcomes. Predicted log quotes are compared with the same log target records; currency MAE is a separate metric. The MUCars honest/attacked biases move upward despite decreasing MAE. The conditional intervals on their paired shifts include zero; this is not evidence of deliberate anchor manipulation. For JUCars $\lambda = 1$, queries cover 10% of all listings,

Table S6: Ten-seed recovery diagnostics: positive-gain violations, raw quote increases and positive increment magnitudes (log units). Median and 90th percentile are zero in these source-level means except synthetic $\lambda = 0$ has mean 90th percentile .004806.

Source	λ	Gain viol. (%)	Quote up (%)	Mean	95th	99th
Synthetic	0	15.79	78.38	0.001648	0.010231	0.030615
Synthetic	1	5.03	79.01	0.000663	0.000345	0.021018
JUCars	0	7.08	86.71	0.001498	0.006725	0.040654
JUCars	1	4.76	87.49	0.001032	0.000248	0.032967
MUCars	0	5.27	87.52	0.000743	0.001096	0.023219
MUCars	1	2.60	83.86	0.000393	0.000000	0.013831

Table S7: Ten-seed mean log quotes and signed log biases at 10% capacity. Queries are per all listings; failures in the last column are per attempted query.

Source	λ	q^H	q^A	Honest bias	Attack bias	Query (%)	Fail (%)
Synthetic	0	10.420905	10.433338	-0.000205	0.012228	10.00	25.1
Synthetic	1	10.416615	10.422780	-0.004495	0.001670	10.00	36.1
JUCars	0	9.044371	9.056807	0.012355	0.024792	10.00	57.3
JUCars	1	9.043418	9.052549	0.011403	0.020534	10.00	61.7
MUCars	0	11.534991	11.541597	0.029095	0.035701	9.03	4.3
MUCars	1	11.540809	11.545842	0.034913	0.039946	10.00	3.1

failures 6.17% and successes 3.83%; conditional failure is 61.7%. A small change in the per-listing failure rate need not imply a small operational failure fraction. Positive log inflation is relative to the model's own honest quote; its percentage reduction is neither a percentage price cut nor realized overpayment avoided. No transaction-price outcome is observed.

Paired budget interaction. Let $D_b = I_{0,b} - I_{1,b}$ at fixed gap .5. We report $D_{.1} - D_0$ without interpreting it as a causal mediation effect: sellers can change their reports between budgets. Intervals independently resample ten common fitting seeds and ten common batches 5,000 times, preserving all weight/budget pairings. These post-hoc marginal percentile intervals are conditional on fixed cohorts and do not cover development-selection uncertainty or guarantee simultaneous coverage. They are distinct from the original studentized simultaneous bands. Zero-query improvements are

Table S8: Budget interaction in positive log inflation. Real-source intervals include zero.

Source	D_0	$D_{.1}$	Interaction	Marginal 95% interval
Synthetic	0.004276	0.006268	0.001992	[0.001255, 0.002713]
JUCars	0.003249	0.003306	0.000057	[-0.000443, 0.000532]
MUCars	0.001527	0.001573	0.000046	[-0.000129, 0.000233]

98.3% and 97.1% of the 10% improvements on JUCars/MUCars. These are descriptive ratios, not causal contribution shares. The evidence supports recovery-graph-inspired regularization with estimator effects; it does not attribute most real-source improvement to verification. For synthetic, changing primary inflation to the fixed-peer unilateral envelope changes the paired reduction from .006267693 to .006266569, a difference of 1.125×10^{-6}. This bounds that mean sensitivity without certifying a joint equilibrium.

S3 Fixed reports versus reselected responses

For each source/seed, retain the $\lambda = 0$, 10%-capacity report profile. Recompute both models' quotes and their own scores/allocations at this fixed profile. Separately evaluate each model's saved, endogenously selected profile. For pre-query quote q_0, highest positive-score query without capacity q_L, hard-capacity quote q_A and honest quote q_H, report signed $E = q_0 - q_H$, $R = q_L - q_0$, $C = q_A - q_L$, $G = q_A - q_H$. All 120 source/seed/weight/profile identities satisfy $E + R + C = G$ to numerical tolerance. A negative mean G can coexist with positive mean $[G]_+$. Fixed reports

Table S9: Ten-seed mean signed decomposition and positive gain, log units. The reference profile is common to the two $\lambda = 0$ report modes.

Source	λ	Profile	E	R	C	G	$[G]_+$
Synthetic	0	Reference	0.005702	-0.002314	0.009044	0.012432	0.012432
Synthetic	1	Fixed report	-0.001008	0.000694	0.003064	0.002749	0.006503
Synthetic	1	Reselected	0.000569	0.000258	0.005338	0.006165	0.006165
JUCars	0	Reference	0.009999	-0.005692	0.008130	0.012437	0.012437
JUCars	1	Fixed report	-0.002822	0.002273	-0.000980	-0.001530	0.006100
JUCars	1	Reselected	0.006706	-0.003770	0.006195	0.009131	0.009131
MUCars	0	Reference	0.002579	-0.000113	0.004140	0.006606	0.006606
MUCars	1	Fixed report	-0.005296	0.004870	-0.001422	-0.001848	0.003343
MUCars	1	Reselected	-0.000309	0.001438	0.003904	0.005033	0.005033

remove report reselection but do not hold the predictor or acquisition head fixed. These are diagnostic identities, not isolated causal interventions. In particular, the real-source R term does not consistently decrease under RPL; a reduction in total positive gain must not be relabeled as a direct reduction of that component.

S4 Matched stronger-gap comparison

Protocol and training-only selection. The fixed protocol precedes the new fits. RPL uses $\lambda \in \{0, .1, .5, 1\}, \beta = .5$; gap-only uses $\beta \in \{.5, 1, 2, 4\}, \lambda = 0$. The common $(0, .5)$ point is shared: seven unique candidates per source/seed, or 210 evaluations. Ninety new gap cells require 360 residual fits (three outer folds and deployment), reusing existing tree backbones. The other 120 candidate cells reuse the ten-seed RPL experiment at 10% capacity. All models use the same no-feedback schedule, folds, minibatch orders, optimizer resets and final OOF head construction. Matching candidate counts and update schedules does not imply equal wall-clock cost. An engineering optimization omits unused intermediate guidance heads in no-feedback training, retaining the final OOF head; a complete JUCars/seed-13 fold replay matched residual parameters, all 1,000-by-11 quotes and training logs exactly.

For each source/seed/family, choose the smallest training OOF mean maximum legal no-query positive log gain, subject to OOF honest currency MAE at most 1.05 times the common $(0, .5)$ reference. Ties use honest MAE then candidate order. With no feasible candidate, choose smallest honest MAE, then gain, then order, and mark constraint failure. The scalar scores and selection receipts are fixed before evaluation; no evaluation metric enters selection. All sixty choices meet the constraint, with no fallback. Gap-only selects $\beta = 4$ thirty times. RPL selects $\lambda = 1$ for ten synthetic, eight JUCars and ten MUCars seeds; JUCars seeds with the other two choices select $\lambda = .5$. OOF feasibility need not imply the same constraint on unseen populations. The original

cohorts had been used in development, so this is not independent confirmation or an expanded-population test.

All candidates and numerical response quality. Table S10 retains the complete grid, including failed queries and finite-response quality. $\rho_{\max}$ is the worst observed regret over all ten seeds and all batches, not a mean of seed maxima; NC is the percentage of all seed–batch responses that did not meet the response criterion. All ninety new gap cells have zero remaining regret and zero nonconverged batches. Reused RPL candidates retain eight nonconverged batches: seven synthetic and one JUCars, with worst regret .023513. Among selected RPL models only two synthetic batches remain nonconverged (2/100, worst .011042); selected real-source RPL responses have zero regret. No failed batch or unfavorable candidate is removed.

Table S10: Complete ten-seed means at 10% capacity. I is positive log inflation; H, A are honest/attacked MAE in source currency; F is failed attempts per all listings (percent), and NC is percent of seed–batches. The shared $(\lambda, \beta) = (0, .5)$ baseline appears once per source.

Source	λ	β	I	H	A	F	$\rho_{\max}$	NC
Synthetic	0	0.5	0.012432	3576.06	3635.45	2.51	0.000000	0.0
Synthetic	0.1	0.5	0.010076	3580.52	3616.78	2.93	0.023513	4.0
Synthetic	0.5	0.5	0.007069	3595.31	3600.02	3.49	0.015078	1.0
Synthetic	1	0.5	0.006165	3605.35	3602.26	3.61	0.011042	2.0
Synthetic	0	1	0.008331	3610.44	3625.83	2.18	0.000000	0.0
Synthetic	0	2	0.005580	3645.26	3652.65	1.40	0.000000	0.0
Synthetic	0	4	0.004035	3676.65	3684.50	1.35	0.000000	0.0
JUCars	0	0.5	0.012437	6966.89	6957.18	5.73	0.000000	0.0
JUCars	0.1	0.5	0.012242	6962.10	6952.73	5.58	0.020806	1.0
JUCars	0.5	0.5	0.010766	6948.32	6941.48	5.35	0.000000	0.0
JUCars	1	0.5	0.009131	6959.75	6956.01	6.17	0.000000	0.0
JUCars	0	1	0.008373	6963.12	6955.09	6.15	0.000000	0.0
JUCars	0	2	0.005721	6903.66	6900.21	4.43	0.000000	0.0
JUCars	0	4	0.003558	6863.28	6860.92	1.84	0.000000	0.0
MUCars	0	0.5	0.006606	27155.74	27306.07	0.43	0.000000	0.0
MUCars	0.1	0.5	0.006408	27038.17	27162.76	0.48	0.000000	0.0
MUCars	0.5	0.5	0.005657	26983.89	27114.40	0.39	0.000000	0.0
MUCars	1	0.5	0.005033	26938.93	27069.81	0.31	0.000000	0.0
MUCars	0	1	0.005012	26897.69	26992.66	0.46	0.000000	0.0
MUCars	0	2	0.004220	26710.03	26808.55	0.46	0.000000	0.0
MUCars	0	4	0.003511	26327.11	26378.61	0.06	0.000000	0.0

Selected comparison and uncertainty. Gap-only has lower positive log inflation in 30/30 paired seeds. Selected means are reported in the main table. Its source-level honest MAE changes relative to selected RPL are $+1.98\%, -1.29\%, -2.27\%$ (ratios of source means). Both honest and attacked MAE improve in 9/10 paired seeds on each real source, but worsen in all ten synthetic seeds. Average failures decrease on all three sources. These are distinct outcomes, not evidence of a scalar welfare optimum. The largest weights are boundary points of the tested grids, not validated optima or recommended defaults.

Table S11 independently resamples ten paired fitting seeds 10,000 times (analysis RNG seed 20261019). Marginal percentile intervals are post-hoc, conditional on fixed records and the chosen

selection rule. They include selection's observed training randomness, but do not cover the development process, new populations or simultaneous multiple comparisons. No independent confirmation claim follows from excluding zero.

Table S11: Paired selected gap-minus-RPL mean differences and marginal 95% intervals. Negative values favor gap-only. Error differences use source currency; inflation differences are log units.

Source	Outcome	Mean difference	Marginal interval
Synthetic	Inflation	-0.002130	[-0.002688, -0.001514]
Synthetic	Honest MAE	71.30	[55.45, 88.12]
Synthetic	Attacked MAE	82.24	[68.44, 95.00]
JUCars	Inflation	-0.006154	[-0.007312, -0.005146]
JUCars	Honest MAE	-89.61	[-131.13, -48.57]
JUCars	Attacked MAE	-86.83	[-126.65, -47.13]
MUCars	Inflation	-0.001521	[-0.002028, -0.001010]
MUCars	Honest MAE	-611.82	[-878.81, -344.71]
MUCars	Attacked MAE	-691.20	[-939.55, -435.54]

Evidence and interpretation. Model, checkpoint, outcome and training-selection hashes bind the comparisons; new-cell inflation and both MAE metrics are independently reaggregated from saved outcomes. All 360 fit-stage records retain the two six-epoch updates and declared gap/path settings. A separate implementation imports no runner and independently checks the candidate sets, constraints and sixty choices from saved OOF scalar scores; it does not regenerate OOF predictions. This is a bounded computational check, not a new complete four-stage audit. The complete grid, scores, selected cells and paired contrasts are released under `notes/review-improvement-v1/gap/`. Scalar receipts are archived separately, excluding weights, arrays and listing-level outcomes.

The comparison does not support practical superiority of RPL over a tuned gap penalty on these cohorts. It also does not test stronger-gap combinations with the five acquisition families: that separate 90-cell crossing holds $\beta = .5$ and tests a different question. The graph loss remains a concrete recovery-targeted formulation; its incremental benefit beyond stronger gap regularization is unestablished under the present selection protocol.

S5 Numerical portability and retained failure

An independently operated Mac completed the original six synthetic/seed-13 cells. Strict numerical reproduction failed under the original tolerances; that verdict is retained. The ten-percent paired absolute improvement is about .008024 locally and .008823 externally (53.97% versus 57.54%). Direction agreement in one seed is supportive, but response quality differs: $\lambda = 1$ maximum regret changes from .0002041 to .0219727 at 10% capacity. The table distinguishes outcome direction from numerical equivalence.

Inputs and tree backbones matched; residual parameters differed. Each machine was internally repeatable in the recorded repeats and thread comparisons. The first recorded difference was an output-layer gradient; hidden forward activations were not captured, so no underlying operator is causally isolated. An isolated float64 candidate is not adopted or validated for scientific performance. We neither relax tolerances nor retroactively substitute a new scientific-equivalence criterion. No new external real-source comparison is claimed. A stronger transportability claim

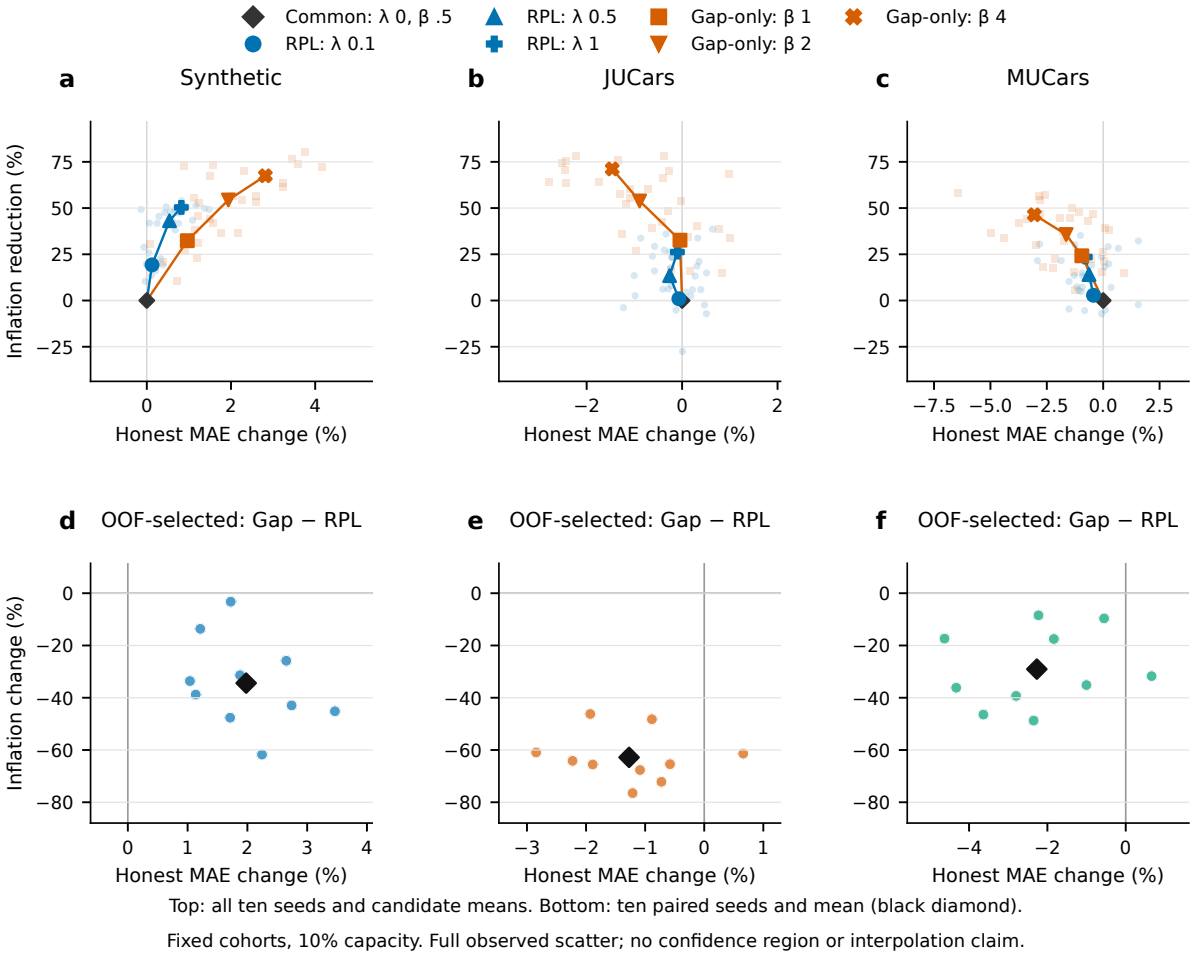


Figure S1: Complete candidate and selected trade-offs at 10% capacity. (a–c) All seven unique candidates over ten fitting seeds. Pale points are individual fits; colored symbols are means of same-seed percentage changes relative to the common $(\lambda, \beta) = (0, .5)$ control. The common point coincides at the origin; connecting lines only guide the candidate order. Higher inflation reduction and lower honest-error change are favorable. (d–f) Ten same-seed percentage changes from selected RPL to selected gap-only; black diamonds are their means. Lower values on both axes are favorable. These means of seed ratios differ from the ratios of source means in the main text. Full observed scatter represents fitting variation, with no confidence region or population extrapolation. Synthetic trades lower inflation for higher error; the real-source means improve on both axes.

Table S12: Existing external-machine numerical comparison, synthetic seed 13. Each entry is local reference / external value. Strict numerical reproduction failed under the original tolerances.

λ	Capacity	Inflation (log)	Honest MAE	Attacked MAE
0	0%	0.0175119/0.0178126	3585.594/3582.048	3720.837/3714.437
0	10%	0.0148686/0.0153343	3585.594/3582.048	3688.439/3683.640
0	30%	0.0128125/0.0131494	3585.594/3582.048	3665.819/3661.133
1	0%	0.0121968/0.0119582	3615.223/3615.545	3676.134/3674.371
1	10%	0.0068443/0.0065113	3615.223/3615.545	3634.360/3621.562
1	30%	0.0041750/0.0041850	3615.223/3615.545	3616.811/3620.643
λ	Capacity	Failed query (%)	Maximum regret (log)	Nonconverged batches /10
0	0%	0.00/0.00	0.0000000/0.0000000	0/0
0	10%	2.70/2.50	0.0000000/0.0000000	0/0
0	30%	5.40/5.40	0.0000000/0.0000000	0/0
1	0%	0.00/0.00	0.0000000/0.0000000	0/0
1	10%	3.80/3.90	0.0002041/0.0219727	1/1
1	30%	7.60/7.40	0.0020809/0.0142298	1/2

requires a future locked protocol and actual independent environment, including effect and regret checks.

The public `reproduction/macos-core-v2` and `reproduction/windows-core-v1` directories contain minimal core training procedures and data routes. The archived software excludes model weights and listing-level outputs; it is not a self-contained binary model package.

S6 Version and evidence map

The immutable software ZIP at 10.5281/zenodo.23447567 is version 1.0.0 from source `89ad56e`. It includes the earlier 90-cell crossing and external difference tables; it excludes manuscripts, large weights, raw data and listing-level outcomes. Later paper revisions and the present diagnostics are not retroactively inside that archive. Current repository citation metadata follows the author order explicitly confirmed by the authors: Tang, You (corresponding), Gu, Pang.

- Pre-review integrated paper: commit `45c2aef`; this is the earlier 28-page manuscript with the five-family crossing in its main text.
- Matched gap runner and engineering replay: commit `5fb3437`; `experiments/run_review_gap_controls_v1.py` and its immutable manifest.
- Mechanisms and cohort construction: commit `00eb579`; `notes/review-improvement-v1/`. Sixty models and 180 existing outcome cells are newly diagnosed, not newly trained.
- Equivalent no-feedback optimization: commit `7d870c2`; `experiments/run_review_gap_controls_v2.py`. The v1 checkpoints remain bound and reusable; the engineering equivalence check precedes continuation.
- Complete stronger-gap analysis and independent selection check: commit `c7a577a`; all 90 new cells, 360 fit records and 210 candidates. The complete trade-off figure and its rendered checks are in `2638dee`.

- Current artifact hashes: `notes/review-improvement-v1/qa.json`. They bind saved files; they do not establish scientific validity.

The main text and this supplement bound claims to the observed fixed populations and declared response procedure. Fitting seeds do not sample new training populations, and independent new-market effectiveness remains unestablished.

S7 Full removals, default scans and seed heterogeneity

The original component experiment toggles suppression scoring (S), feedback (F) and gap loss (G). All eight configurations retain backbone, augmentation, error and anchoring terms, with equal role separation and epoch budgets. Thus removal of two or all three components is evaluated against a trained baseline (Figure S2). These switches do not constitute three independently established innovations.

Figure S3 retains all ten fitting seeds and four nonzero path/gap comparisons, each with its same-gap zero-path reference. Negative seed effects at smaller weights remain. A separate original *three-seed* recovery-edge pilot found mean positive increments decreasing from .001790 to .000693 (synthetic), .001648 to .001048 (JUCars) and .000788 to .000450 (MUCars) at $\lambda = 1, \beta = .5$. All remained positive: zero path loss was not achieved, and the pilot is not relabeled as ten-seed evidence.

Core scans vary gap weight $0, .1, .25, .5, 1, 2$, tolerance $0, .01, .02, .05, .1$ and feedback rounds $1, 2, 4$, at 10/30% budgets. Implementation scans vary anchoring, attack weight, learning rate, epochs, head trees and width. Figures S4 and S5 explicitly mark every scanned value and engineering default; matching default observations are reused exactly, not retrained or selected on evaluation outcomes. These three-seed scans do not retune ten-seed RPL or prove an optimum. The separate availability stress adds 0/10/30% natural absence to fixed models (189 cells, 63 reused), with seven policies and fresh response search but no new fitting; it is a controlled perturbation rather than a new market or actual-seller validation.

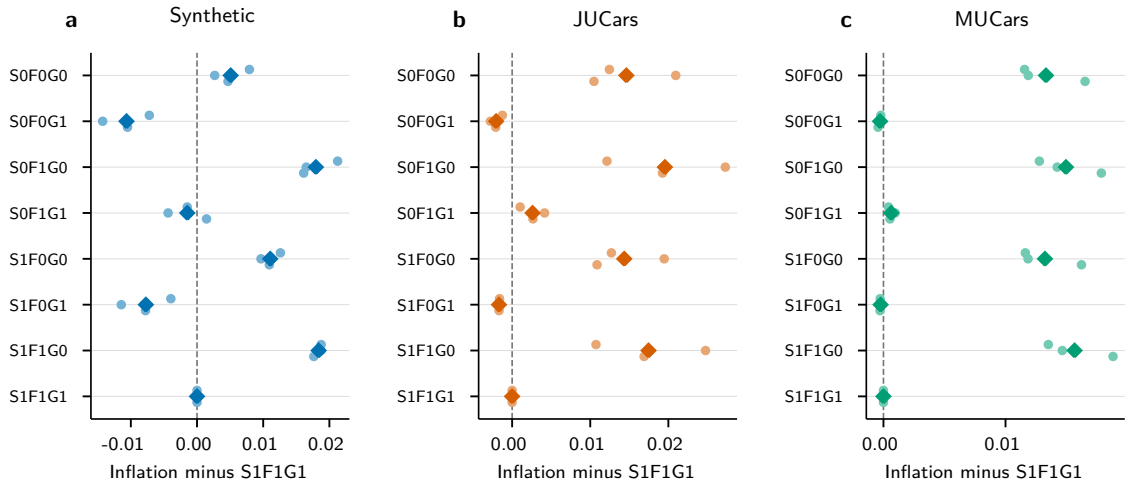
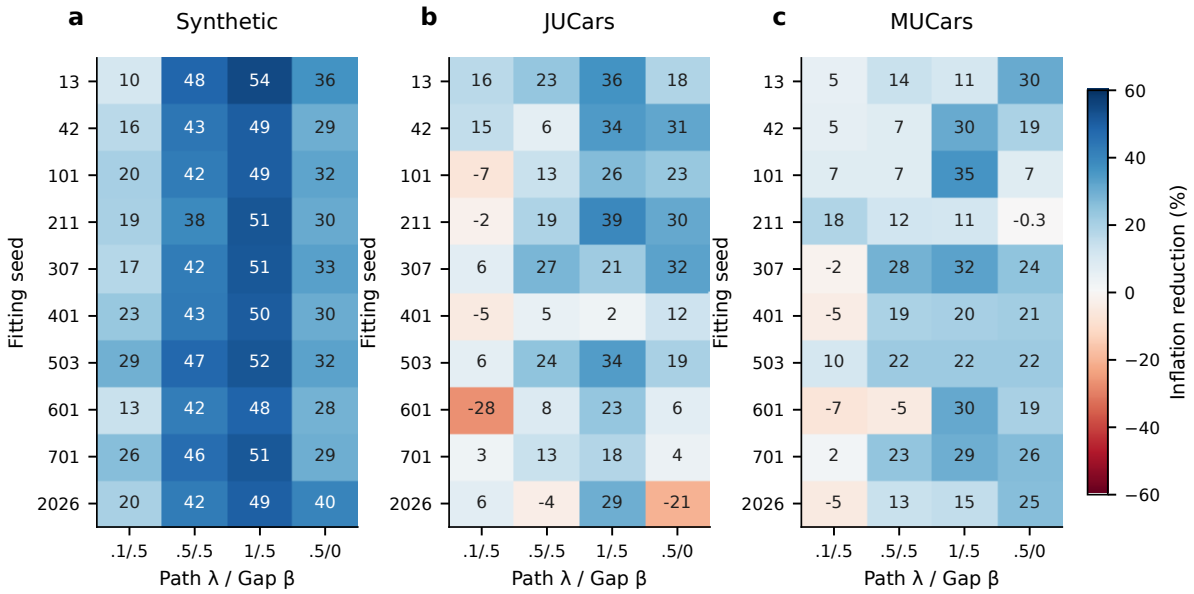


Figure S2: Complete three-component ablation at 10% capacity: circles show three paired seed differences from S1F1G1, diamonds their means. Negative values favor the ablated configuration. S/F/G denote suppression, feedback and gap switches; nonconverged responses remain.



10% capacity. Each cell is a paired seed ratio against its same-gap zero-path control.
Positive: lower inflation; negative: higher. No seed or nonconverged result is removed.

Figure S3: Ten-seed heterogeneity at 10% capacity. Positive values mean lower inflation than same-gap zero-path controls. Negative effects and nonconverged responses are retained. Labels are rounded; exact values are in the public source CSV.

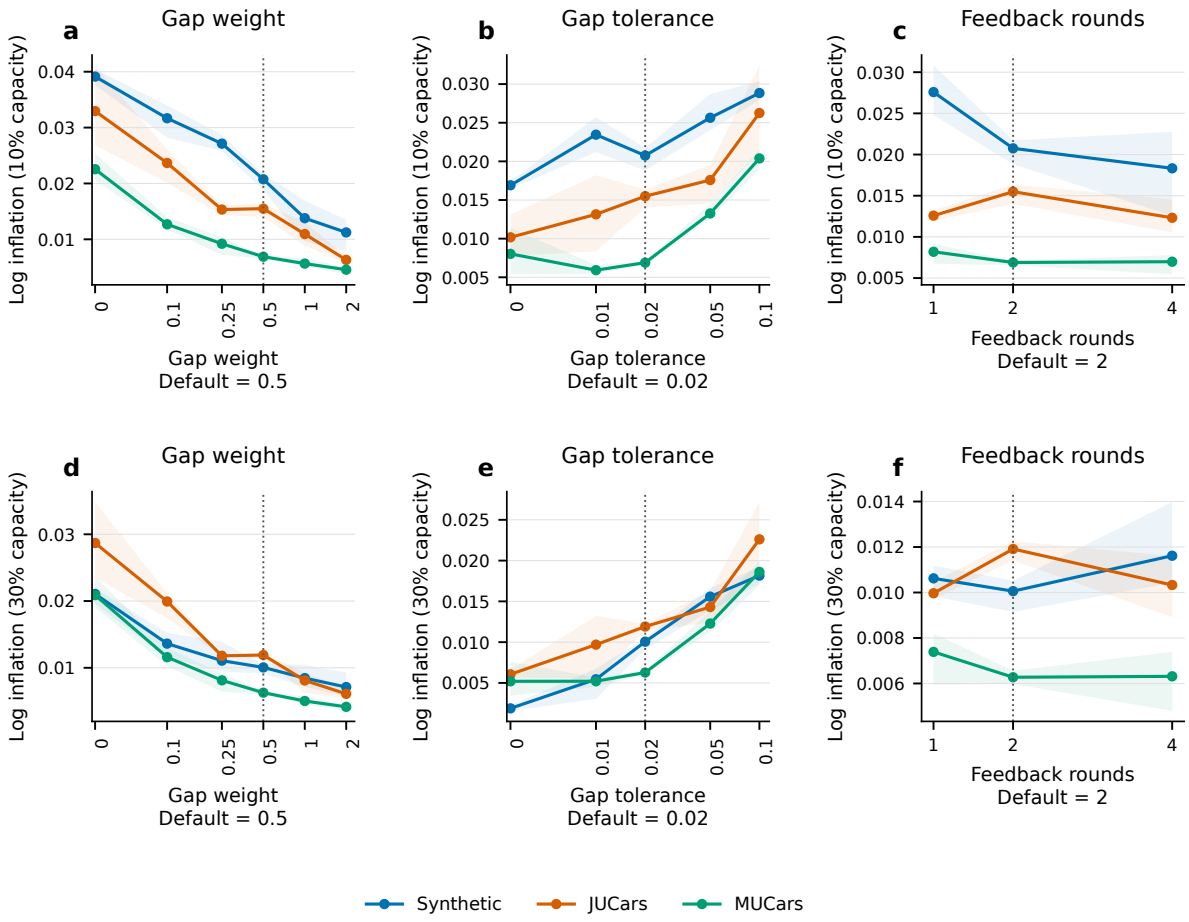


Figure S4: Core-parameter scans in the original feedback implementation, with three fitting seeds. Dotted lines mark design defaults and all evaluated settings are ticked. Gap weight and tolerance use symmetric-log coordinates. Means and full ranges are descriptive, not confidence intervals. The defaults are engineering choices, not evaluation-selected optima.

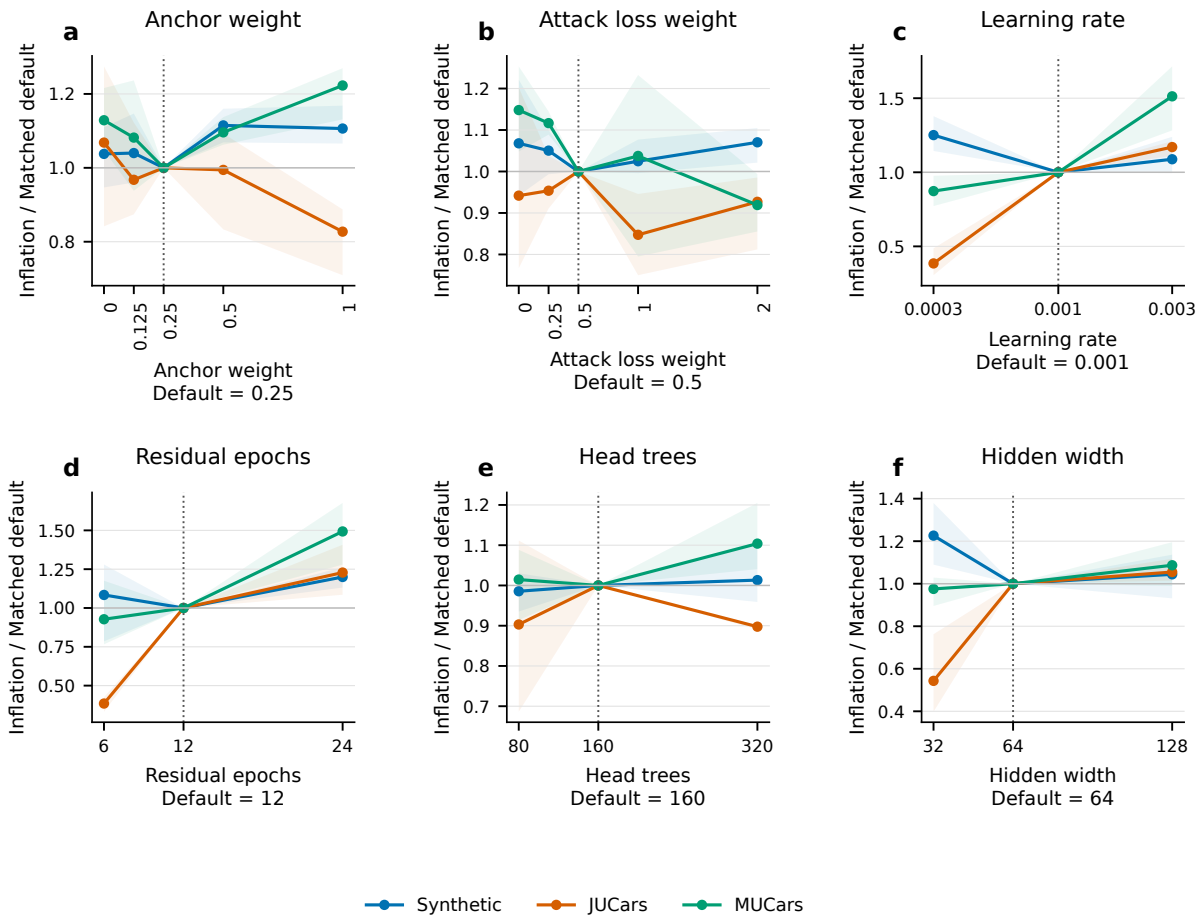


Figure S5: Six implementation-parameter scans at 10% capacity, retaining three fitting seeds. Every ratio uses the same-seed matched default. Dotted lines mark defaults; ticks show all evaluated values, and learning rate uses a logarithmic axis. Curves show means and full seed ranges, not confidence intervals.